Venkata Lakshmi Parimala Pasupuleti

AI Evaluation & Feedback Platform Engineer · Python / LangGraph / LLMOps · HITL Workflows · Healthcare & Regulated-Domain AI

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SUMMARY

AI Evaluation and Feedback Platform Engineer with 9 years of production software ownership. Built AgentProbe -- an open-source evaluation platform for production AI agents: CI-integrated regression gate (failures caught before deployment, not discovered in production), LLM-as-judge scoring with agent tracing (every failure has a diagnosable root cause), and a multi-trial pass@k harness (human feedback measurably changes agent behavior over time). Built LeaseI -- an 8-agent LangGraph system with HITL escalation on regulated decisions, evaluated across 1,240 test cases. Regulated-domain healthcare AI experience: AI on FHIR (HIPAA-compliant PHI system, RBAC) + Entrata PHI compliance (zero security incidents). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, 2026).

SKILLS

AI Evaluation & LLMOps: LLM Evaluation, Evaluation Pipelines, Feedback Systems, Regression Suites, CI Eval Gate, LLM-as-Judge, Pass@k Harness, Agent Observability, Agent Tracing, Production Monitoring, Hallucination Detection, Quality Metrics, Root Cause Analysis, HITL (Human-in-the-Loop), Guardrails, Output Validation, Confidence Scoring, Failure Diagnostics, Reliability Engineering, ML Pipelines

AI & Agents: LangGraph, LangChain, Agentic AI, Conversational AI, Multi-Agent Orchestration, LLM Orchestration, RAG Pipelines, ChromaDB, Vector Database, Semantic Search, Embeddings, Prompt Engineering, Tool Calling, OpenAI API, Anthropic API, Groq, Ollama, MCP

Languages & Frameworks: Python, JavaScript, TypeScript, Java, SQL, PL/SQL, FastAPI, REST APIs, React.js, Flask, Angular, Node.js, Bash, HTML/CSS

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Docker, Kubernetes (K8s), CI/CD (GitHub Actions), RabbitMQ, Redis, Microservices, HIPAA Compliance, PHI Data Handling, PostgreSQL, MySQL, Oracle, Git, Linux, Agile/Scrum, Libsodium

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern

- Built a production LangGraph agentic system with RAG retrieval, guardrails, and failure handling for an external client -- identified and fixed quality failures across reasoning and retrieval layers before delivery; every compliance decision cited to exact source regulation.

Entrata (PropTech SaaS · 100K+ MAU · PHI compliance · regulated-domain production systems)
Sep 2018 - Aug 2024

Senior Software Engineer

- **Owned compliance-critical data systems in a regulated domain where agent failures have real consequences** -- annual HUD income-limit pipeline (180,000 domain records determining housing eligibility for 100K+ residents); zero compliance violations across 7 years. Every release included baseline validation, reconciliation, and rollback preparation -- because a wrong output means a wrong eligibility decision for thousands of people.
- Led AWS S3 document migration -- 100% PHI compliance with Libsodium-encrypted pipelines, zero security incidents across the full resident data migration; designed and owned HIPAA-adjacent data workflows throughout tenure.
- Shipped 20+ full-stack production modules (Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints; built Python/Bash automation pipelines eliminating 80% of manual operational workload; optimized 200+ SQL queries (30% performance gain).
- Mentored 5+ junior engineers; conducted 300+ peer code reviews; built internal debugging tools reducing investigation time by 40%.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built multi-source prediction models (Python, R) -- $R^2 = 0.977$; applied Isolation Forest for automated data integrity checks across 3 research pipelines; delivered Python + database workshops to 600+ students.

PROJECTS

AgentProbe: AI Evaluation & Feedback Platform for Production Agents
2025 | github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions)

- Built an evaluation and feedback platform for production AI agents: CI-integrated regression gate (failures detected by systems before deployment), LLM-as-judge scoring with full agent tracing (every failure has a diagnosable root cause and scored rationale), multi-trial pass@k harness measuring whether changes actually improve agent behavior (human feedback loop with measurable lag reduction over time).
- 5-metric eval engine (correctness, groundedness, hallucination rate, latency, token efficiency); multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama); 15-endpoint FastAPI backend; evaluated LeaseI across 1,240+ production-quality test cases.

Leasel: 8-Agent LangGraph Platform with HITL on Regulated Decisions
2026 | github.com/venkatalakshmiparimala/Lease_complaince_rag

Python · LangGraph (8-agent) · FastAPI · RAG · OCR · SQLite · Streamlit

- Built HITL escalation as a structural constraint: when agent confidence drops below threshold or the decision is regulated, the system routes to human review and captures feedback -- the architecture that transforms human review into measurable agent improvement.
- Diagnosed and categorized failures across reasoning, retrieval, and orchestration layers; 93.6% precision across 1,240 compliance test cases with full observability through AgentProbe.

AI on FHIR: HIPAA-Compliant Medical Records Query System

2024 | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask · React · LLMs · FHIR · spaCy · BERT · Docker

- HIPAA-compliant AI system over structured PHI data -- RBAC access controls, NLP-to-FHIR API translation, LLM + BERT integration; reduced manual clinical retrieval time ~60%. Healthcare regulated domain in production.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — Graduated May 2026 | GPA: 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017